



Tecnológico de Monterrey

Campus Querétaro

Laboratorio: Funciones agregadas y sub-consultas

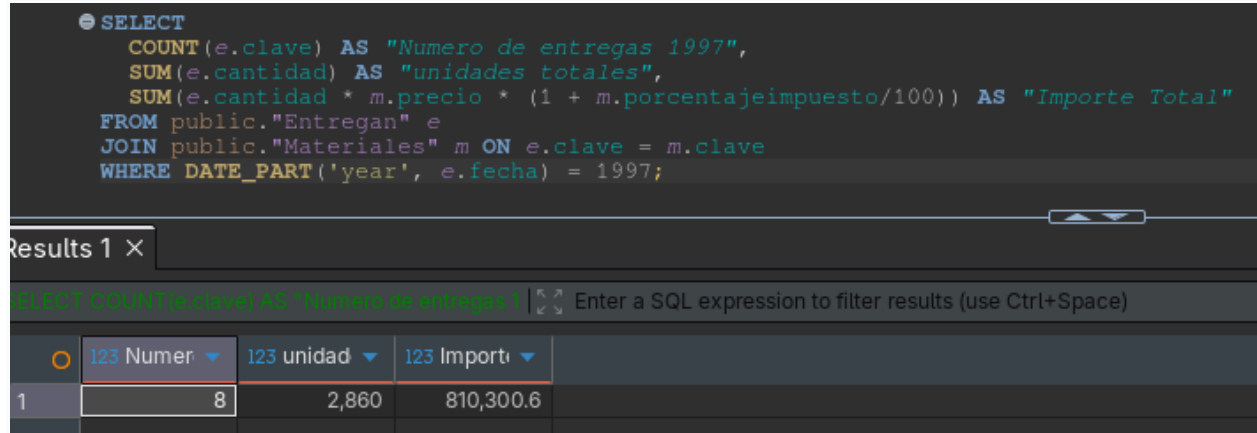
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Materia: Construcción de software y toma de decisiones (Gpo 501)

Fecha de entrega
17 de Mayo 2026

Primera Consulta:

```
SELECT
  COUNT(e.clave) AS "Numero de entregas 1997",
  SUM(e.cantidad) AS "unidades totales",
  SUM(e.cantidad * m.precio * (1 + m.porcentajeimpuesto/100)) AS "Importe Total"
FROM public."Entregan" e
JOIN public."Materiales" m ON e.clave = m.clave
WHERE DATE_PART('year', e.fecha) = 1997;
```

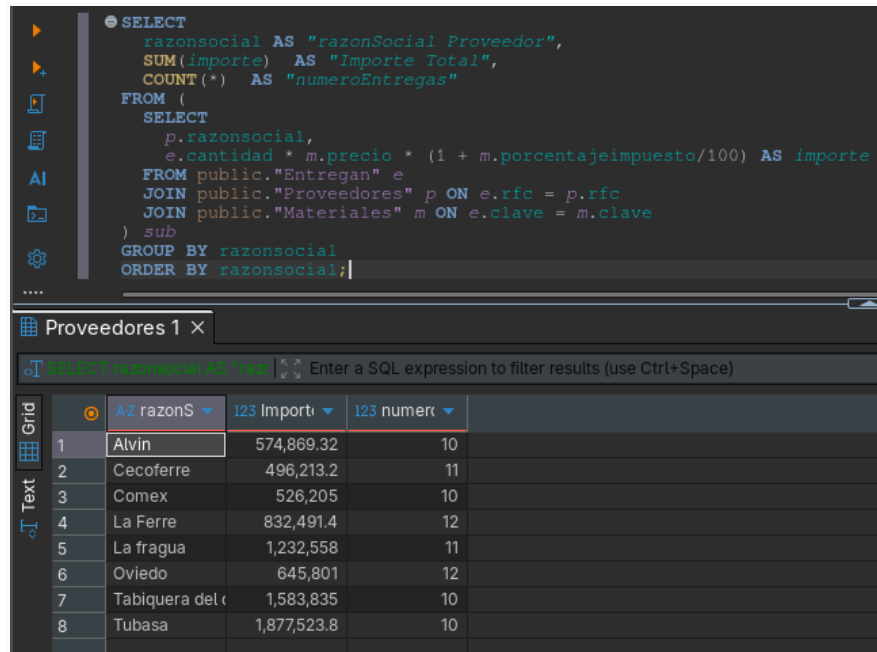


The screenshot shows a SQL query editor with the first query pasted in. Below the query, the results are displayed in a table with 4 columns: 'Numero de entregas 1997', 'unidades totales', 'Importe Total', and an empty column. The first row of results shows the values 8, 2,860, and 810,300.6.

	123 Numer	123 unidad	123 Importe	
1	8	2,860	810,300.6	

Segunda Consulta:

```
SELECT
  razonsocial AS "razonSocial Proveedor",
  SUM(importe) AS "Importe Total",
  COUNT(*) AS "numeroEntregas"
FROM (
  SELECT
    p.razonsocial,
    e.cantidad * m.precio * (1 + m.porcentajeimpuesto/100) AS importe
  FROM public."Entregan" e
  JOIN public."Proveedores" p ON e.rfc = p.rfc
  JOIN public."Materiales" m ON e.clave = m.clave
) sub
GROUP BY razonsocial
ORDER BY razonsocial;
```



```

SELECT
    razonsocial AS "razonSocial Proveedor",
    SUM(importe) AS "Importe Total",
    COUNT(*) AS "numeroEntregas"
FROM (
    SELECT
        p.razonsocial,
        e.cantidad * m.precio * (1 + m.porcentajeimpuesto/100) AS importe
    FROM public."Entregan" e
    JOIN public."Proveedores" p ON e.rfc = p.rfc
    JOIN public."Materiales" m ON e.clave = m.clave
) sub
GROUP BY razonsocial
ORDER BY razonsocial;

```

	AZ razonS	123 Importe	123 numer
1	Alvin	574,869.32	10
2	Cecoferre	496,213.2	11
3	Comex	526,205	10
4	La Ferre	832,491.4	12
5	La fragua	1,232,558	11
6	Oviedo	645,801	12
7	Tabiquera del c	1,583,835	10
8	Tubasa	1,877,523.8	10

Tercera consulta:

```

SELECT
    sub.clave AS "Clave",
    sub.descripcion AS "descripcion",
    sub.cantidad_total AS "cantidad total",
    sub.max_cantidad AS "maxcantidad",
    sub.min_cantidad AS "minCantidad",
    sub.importe_total AS "Importe Total"
FROM (
    SELECT
        m.clave,
        m.descripcion,
        SUM(e.cantidad) AS cantidad_total,
        MAX(e.cantidad) AS max_cantidad,
        MIN(e.cantidad) AS min_cantidad,
        AVG(e.cantidad) AS avg_cantidad,
        SUM(e.cantidad * m.precio * (1 + m.porcentajeimpuesto/100)) AS importe_total
    FROM public."Entregan" e
    JOIN public."Materiales" m ON e.clave = m.clave
    GROUP BY m.clave, m.descripcion
) sub
WHERE sub.avg_cantidad > 400
ORDER BY sub.clave;

```

```

SELECT
  sub.clave AS "Clave",
  sub.descripcion AS "descripcion",
  sub.cantidad_total AS "cantidad total",
  sub.max_cantidad AS "maxcantidad",
  sub.min_cantidad AS "minCantidad",
  sub.importe_total AS "importe Total"
FROM (
  SELECT
    m.clave,
    m.descripcion,
    SUM(e.cantidad) AS cantidad_total,
    MAX(e.cantidad) AS max_cantidad,
    MIN(e.cantidad) AS min_cantidad,
    AVG(e.cantidad) AS avg_cantidad,
    SUM(e.cantidad * m.precio * (1 + m.porcentajeimpuesto/100)) AS importe_total
  FROM public."Entregan" e
  JOIN public."Materiales" m ON e.clave = m.clave
  GROUP BY m.clave, m.descripcion
) sub
WHERE sub.avg_cantidad > 400
ORDER BY sub.clave;

```

Materiales 1 X

SELECT sub.clave AS "Clave", sub.descripcion AS "descripcion" Enter a SQL expression to filter results (use Ctrl+Space)

	123 Clave	A2 descri	123 cantida	123 maxca	123 minCai	123 Importe
1	1,010	Varilla 4/32	1,051	528	523	140,203.4
2	1,040	Varilla 3/18	803	540	263	149,036.8
3	1,050	Varilla 4/34	1,126	623	503	228,578
4	1,060	Varilla 3/19	1,016	692	324	223,926.4
5	1,100	Block	1,165	699	466	40,542
6	1,120	Sillar rosa	907	692	215	105,212
7	1,140	Cantera blanc	802	583	219	186,064
8	1,150	Cantera gris	911	458	453	1,278,679.6
9	1,230	Cemento	842	530	312	293,016
10	1,260	Gravilla	1,091	631	460	113,900.4
11	1,270	Tezontle	1,052	546	506	97,625.6
12	1,320	Tubería 4.4	1,111	698	413	296,414.8
13	1,340	Tubería 4.5	998	674	324	289,420
14	1,380	Pintura C1011	947	645	302	796,427
15	1,390	Pintura B1021	804	697	107	116,580
16	1,410	Pintura B1021	1,068	601	467	154,860
17	1,420	Pintura C1012	881	603	278	127,745

Cuarta Consulta:

```

SELECT
  sub.razonsocial AS "RazonSocial",
  sub.promedio AS "PromedioCantidad",
  sub.clave AS "ClaveMaterial",
  sub.descripcion AS "descripcionMaterial"
FROM (
  SELECT
    p.razonsocial,
    m.clave,
    m.descripcion,
    AVG(e.cantidad) AS promedio
  FROM public."Entregan" e
  JOIN public."Proveedores" p ON p.rfc = e.rfc
  JOIN public."Materiales" m ON e.clave = m.clave
  GROUP BY p.rfc, p.razonsocial, m.clave, m.descripcion
) sub
WHERE sub.promedio >= 500
ORDER BY sub.clave, sub.descripcion;

```

```

SELECT
    sub.razonsocial AS "RazonSocial",
    sub.promedio AS "PromedioCantidad",
    sub.clave AS "ClaveMaterial",
    sub.descripcion AS "descripcionMaterial"
FROM (
    SELECT
        p.razonsocial,
        m.clave,
        m.descripcion,
        AVG(e.cantidad) AS promedio
    FROM public."Entregan" e
    JOIN public."Proveedores" p ON p.rfc = e.rfc
    JOIN public."Materiales" m ON e.clave = m.clave
    GROUP BY p.rfc, p.razonsocial, m.clave, m.descripcion
) sub
WHERE sub.promedio >= 500
ORDER BY sub.clave, sub.descripcion;

```

Proveedores(+) 1 X

SELECT sub.razonsocial AS "RazonSocial", sub.promedio AS "PromedioCantidad", sub.clave AS "ClaveMaterial", sub.descripcion AS "descripcionMaterial" Enter a SQL expression to filter results (use Ctrl+Space)

	AZ RazonSocial	123 PromedioCantidad	123 ClaveMaterial	AZ descripcionMaterial
1	Oviedo	525.5	1,010	Varilla 4/32
2	Comex	563	1,050	Varilla 4/34
3	Tabiquera del centro	508	1,060	Varilla 3/19
4	La Ferre	582.5	1,100	Block
5	La Ferre	545.5	1,260	Gravilla
6	Cecoferre	526	1,270	Tezontle
7	La fragua	555.5	1,320	Tubería 4.4
8	Oviedo	534	1,410	Pintura B1021

Quinta consulta:

```

SELECT *
FROM (
    SELECT
        p.razonsocial AS "Proveedor",
        m.clave AS "Clave Mat",
        m.descripcion AS "Material",
        AVG(e.cantidad) AS "Cantidad Promedio"
    FROM public."Entregan" e
    JOIN public."Proveedores" p ON e.rfc = p.rfc
    JOIN public."Materiales" m ON e.clave = m.clave
    GROUP BY p.rfc, p.razonsocial, m.clave, m.descripcion
) AS promedios
WHERE "Cantidad Promedio" < 370
OR "Cantidad Promedio" > 450
ORDER BY "Cantidad Promedio";

```

Proveedores(+) 1 x

 `SELECT * FROM ( SELECT p.razonsocial AS "Prover`  Enter a SQL expression to filter results (use Ctrl+Space)

Refresh Save Cancel : : Export data 200 35